

About IGA™ — Intelligence Governance Architecture

Artifact:

IGA™ — Intelligence Governance Architecture

Artifact Type:

OOF® Master Governance Reference Map

Official Map:

IGA™ Master Governance Space Map

Domain:

Intelligence Governance

Scope:

AI Systems · Autonomous Agents · Multi-Agent Systems · Robotics · Human-AI Environments · Intelligent Infrastructure · Future Intelligent Systems

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OOF® — OriginOpen® Foundation

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1. What Is IGA™?

IGA™ — Intelligence Governance Architecture is the OOF® master governance reference map for identifying, distinguishing, navigating, and relating the primary governance spaces of intelligent and autonomous systems.

IGA™ does not define a specific AI model, technology, vendor, platform, implementation, or system architecture.

It provides a common governance topology for determining where a governance problem belongs before deeper governance begins.

IGA™ therefore operates as a diagnostic and navigational reference layer across intelligent systems.

Its purpose is not to force every governance problem into a single framework.

Its purpose is to distinguish the fundamental governance spaces involved, establish their boundaries, and show how those spaces may interact without becoming the same governed domain.

2 Why Does IGA™ Exist?



Intelligent systems increasingly combine computation, cognition, memory, agency, external execution, autonomous operation, interaction with Operational Reality, evidence generation, attribution, and accountability.

When a governance problem occurs, identifying the correct governed space becomes essential.

A problem may originate in identity rather than cognition.

A system may reason correctly while acting without sufficient authority or permission.

An authorized action may produce an unintended operational outcome.

What actually happened may differ from what was observed.

An observation may exist without becoming governance-relevant evidence.

Evidence may exist without being sufficiently validated.

An event may be attributable without automatically resolving accountability.

A system may also cross several governance spaces during one activity while the primary governance ownership of a specific problem remains elsewhere.

IGA™ provides the reference structure needed to make these distinctions explicit.

3. Core Reference Question

IGA™ answers the foundational question:

Where does the governance problem belong?

This question precedes deeper architecture-specific governance.

IGA™ identifies the relevant governance space.

The applicable OOF® Governance Architecture, governance mechanism, or domain-specific framework can then determine how that space should be governed.

4. The IGA™ Master Governance Space Map

IGA™ identifies fourteen primary governance spaces:

01 — Governance Who establishes governance and how does it function?

02 — Identity Who or what is acting?

03 — Computation How is intelligent activity computationally produced?

04 — Cognition How does the system think?

05 — Memory What does the system remember?

06 — Agency Who or what can form intent and pursue goals?

07 — Execution What activity is actually initiated or performed?

08 — Operational Reality What actually happened?

09 — Observation What was actually observed?

10 — Evidence What governance-relevant evidence exists?

11 — Validation Can it be established as sufficiently valid and reliable?

12 — Attribution To what can it be attributed?

13 — Accountability Who bears responsibility?

14 — Auditability

Can it be reconstructed?

Together, these spaces form the core intelligence governance topology of IGA™.

5. Governance-Space Topology — Not a Mandatory Lifecycle

The numbered arrangement of IGA™ is a reference structure.

It does not require every intelligent activity to mechanically traverse all fourteen spaces in sequence.

Governance spaces may operate simultaneously, recursively, conditionally, or independently.

An activity may terminate before execution.

Memory may influence cognition repeatedly.

Governance may intervene before, during, or after execution.

Observation may be incomplete.

Attribution may remain unresolved.

Auditability may be required across multiple stages of activity rather than only after an event.

IGA™ therefore defines governance-space topology, not a mandatory lifecycle.

Lifecycle governance of intelligent activity is addressed separately through GIARM™ — Governed Intelligent Activity Reference Model.

6. Governance

Governance establishes the conditions under which governance itself exists and functions.

This space concerns the legitimate structures through which authority, scope, boundaries, delegation, accountability, validation, integrity, transparency, continuity, and interoperability may be established and maintained.

Its foundational question is:

Who establishes governance and how does it function?

Governance provides the governing context within which the remaining IGA™ spaces operate.

7. Identity

Identity determines who or what is acting.

It establishes the basis through which actors, systems, agents, organizations, operational personas, or other governed entities can be distinguished, traced, and related to legitimate origin and continuity.

Identity is foundational to authority, delegation, attribution, responsibility, and accountability.

A capability to create, assume, or present an identity does not itself establish legitimate authority to represent that identity.

Identity Capability ≠ Identity Authority

8. Computation

Computation governs the space in which intelligent activity is computationally produced.

It concerns the computational processes and mechanisms through which system states, outputs, or intelligent capabilities emerge.

Computation is intentionally technology-neutral.

IGAT™ does not assume that future intelligence must depend exclusively on neural networks or any current computational paradigm.

This creates an important distinction:

Computation ≠ Cognition

Computation concerns how processing occurs.

Cognition concerns how information becomes interpretation, reasoning, understanding, or decision formation.

9. Cognition

Cognition concerns interpretation, reasoning, understanding, contextualization, reality modelling, and cognitive decision formation.

It asks:

How does the system think?

Cognition is distinct from computation, memory, agency, and execution.

A system may computationally process information without producing meaningful cognition.

It may form a cognitive conclusion without storing it.

It may reason without possessing authority to act.

And it may form a decision without that decision ever becoming an executed action.

10. Memory

Memory governs what persists beyond immediate processing and what may subsequently influence cognition, decisions, agency, or activity.

It includes memory formation, persistence, retrieval, provenance, continuity, modification, and governed use.

Memory is not simply stored information.

Governance must be capable of distinguishing what persists, why it persists, where it originated, how it changed, and under what conditions it may influence future intelligent activity.

11. Agency

Agency concerns the capacity of a system or actor to form intent, pursue goals, plan activity, coordinate actions, use tools, or delegate activity.

Its foundational question is:

Who or what can form intent and pursue goals?

Agency must remain distinct from cognition and execution.

Cognition ≠ Agency

Reasoning about an action does not itself establish agency.

And:

Agency ≠ Execution

Possessing goals, plans, or action-selection capability does not establish that an external action occurred.

12. Execution

Execution concerns activity that is actually initiated or performed.

It marks the governance space in which intended, selected, delegated, or authorized activity becomes execution.

Execution must remain distinguishable from capability, authority, permission, decision formation, and operational outcome.

A system being capable of performing an action does not establish legitimate authority to perform it.

Authority does not automatically establish permission for every specific action.

A decision does not establish that execution occurred.

And execution does not establish the resulting Operational Reality.

13. Operational Reality

Operational Reality establishes what actually happened.

It concerns actual states, events, interactions, effects, consequences, outputs, and outcomes produced within operational environments.

Operational Reality remains independent of what a system intended, predicted, reported, or believed would occur.

This establishes one of the foundational boundaries of intelligent-system governance:

Action $\neq$ Outcome

and:

Reality $\neq$ Observation

Operational Reality remains the reference point against which claims about intelligent activity must ultimately be examined.

14. Observation

Observation concerns what was actually detected, measured, recorded, reported, or otherwise perceived about Operational Reality.

Observation is necessary for governance, but it is not equivalent to reality itself.

Sensors may fail.

Logs may be incomplete.

Agents may report incorrectly.

Human observers may possess only partial information.

Different systems may observe different parts of the same event.

Therefore:

Reality $\neq$ Observation

Observation provides access to Operational Reality but does not replace it.

15. Evidence

Evidence concerns governance-relevant data, records, traces, artifacts, observations, or other material capable of supporting governance examination.

Not every observation automatically becomes governance-relevant evidence.

Evidence requires sufficient relationship to the governance question being examined.

Therefore:

Observation ≠ Evidence

IGA™ distinguishes the existence of observations from the existence of evidence capable of supporting governance decisions.

16. Validation

Validation determines whether relevant claims, states, observations, evidence, or conditions can be established as sufficiently valid and reliable for the governance decision being made.

Evidence does not validate itself.

Its provenance, authenticity, relevance, reliability, completeness, context, and other applicable conditions may require examination.

Therefore:

Evidence ≠ Validation

Validation provides the governance basis for determining what may responsibly be relied upon.

17. Attribution

Attribution concerns whether relevant activity, decisions, actions, events, effects, or consequences can be connected to identifiable actors, systems, agents, authorities, or other governed entities.

Attribution must not be forced where the available evidence cannot establish it.

IGA™ therefore treats attribution as:

Attribution — Where Establishable

Unresolved attribution is itself a governance state that may require documentation, further investigation, additional evidence, or continued reassessment.

18. Accountability

Accountability determines who bears what governance responsibility for the relevant governed activity, decision, authority, action, system relationship, or consequence.

Attribution and accountability are related but not interchangeable.

Knowing that a system performed an action does not automatically determine the complete accountability structure surrounding that action.

Accountability may involve system operators, delegating authorities, organizations, governance owners, decision-makers, infrastructure providers, or other actors depending on the governed relationship.

Therefore:

Attribution ≠ Accountability

19. Auditability

Auditability determines whether relevant intelligent activity and its governance can be reconstructed, examined, traced, and audited.

Its core question is:

Can it be reconstructed?

Auditability may depend on evidence, provenance, identity continuity, decision records, authority records, execution traces, Operational Reality observations, validation outputs, and attribution information.

Within the current OOF® architecture system, this space is associated with:

ADIT® — Audit & Evidence Governance Architecture Auditability is not merely an end-stage function.

It may need to exist throughout governed intelligent activity.

20. Cross-Cutting Governance Dimensions

Not every fundamental governance requirement is a separate governance space.

Some conditions operate across multiple IGA™ spaces.

The Master Governance Space Map identifies the following principal cross-cutting dimensions:

Authority — legitimate source and extent of governance authority.

Permission — what specific activity is permitted under applicable authority and conditions.

Boundaries — where governance, authority, activity, or responsibility begins and ends.

Responsibility — governance obligations associated with actors, roles, decisions, and activities.

Integrity — preservation of relevant governance, system, information, and operational integrity.

Safety — governance of conditions necessary to prevent or control unacceptable harm.

Human Override — legitimate human capability to intervene, restrict, suspend, or terminate activity where required.

Interoperability — ability of governance structures to operate coherently across systems and governance domains.

Transparency — sufficient visibility into relevant governance structures, conditions, decisions, or activity.

Continuity — preservation of governance across time, transitions, system changes, and operational evolution.

These dimensions may affect several governance spaces without becoming identical to those spaces.

21. Domain-Specific Governed Spaces

IGA™ distinguishes universal governance spaces from specialized governed domains.

Intelligent activity may generate consequences or governance questions involving:

Value Value creation, movement, transformation, preservation, or loss.

Harm Harm prevention, occurrence, propagation, impact, and response.

Tax Tax-relevant events and their governance.

Liability Legal and operational liability relationships.

Risk Risk formation, exposure, control, and management.

Energy Energy and resource use associated with intelligent activity.

Environment Environmental effects and consequences.

Simulation Simulated environments, simulated authority, and boundaries between simulation and Operational Reality.

Sector-Specific Governance Governed domains arising from particular industries, infrastructures, professions, or operational environments.

Future Governance Spaces New governed domains discovered as intelligent systems and their operational environments evolve.

These spaces are not mandatory stages of every intelligent activity.

They connect to IGA™ where the relevant governed object or consequence arises.

22. Cross-Space Governance

A governance problem may involve several IGA™ spaces simultaneously.

For example, an autonomous agent may involve:

Identity to establish who or what is acting.

Agency to establish goal-directed activity.

Authority and Permission to determine whether the activity may proceed.

Execution to determine what action was performed.

Operational Reality to establish what actually happened.

Observation and Evidence to establish what can be examined.

Validation to determine what may reliably be relied upon.

Attribution to establish relevant causal or operational relationships where possible.

Accountability to determine governance responsibility.

Auditability to reconstruct and examine the governed activity.

The involvement of several spaces does not eliminate their boundaries.

23. Cross-Architecture Ownership Principle

IGA™ follows a fundamental OOF® boundary principle:

An operational mechanism may produce inputs for multiple governance architectures, but it belongs only to the architecture whose governed domain includes that mechanism itself. Other architectures govern only its domain-specific relevance or consequences.

Accordingly, cross-space interaction does not automatically transfer governance ownership.

A governance problem may cross multiple spaces while retaining a primary governed domain.

This prevents overlapping architectures from becoming duplicative governance systems.

24. Architectural Principles

IGA™ establishes four primary architectural principles:

1. Each governance space answers a distinct primary governance question.
2. Governance spaces may interact without becoming the same governed domain.
3. A governance problem may cross multiple spaces while retaining primary governance ownership.
4. Together, the spaces form the complete intelligence governance reference map.

These principles allow IGA™ to expand without requiring every newly discovered governance concern to become a new universal layer.

25. Relationship to GIARM™

IGA™ and GIARM™ address different governance questions.

IGA™ asks:

Where does the governance problem belong?

GIARM™ asks:

How does intelligent activity remain governable from purpose and authority through decision, action, Operational Reality, evidence, and governance adaptation?

IGA™ is therefore a governance-space reference map.

GIARM™ is an end-to-end governance lifecycle reference model.

They are complementary, not interchangeable.

26. Relationship to Layers of Governed Intelligence

Layers of Governed Intelligence identifies the functional layers through which intelligent activity occurs:

Intelligence · Cognition · Memory · Agency · Execution · Operational Reality IGA™ performs a different function.

It identifies the governance spaces necessary to diagnose where governance questions, failures, evidence, responsibility, validation, and auditability belong.

In simplified form:

Layers of Governed Intelligence → Where intelligent activity occurs.

IGA™ → Where the governance problem belongs.

27. Relationship to the Governable Intelligent Activity Decision Tree™

The Governable Intelligent Activity Decision Tree™ determines:

Under what governance conditions may intelligent activity proceed?

IGA™ does not make that decision.

Instead, it provides the governance-space topology necessary to identify which governed domains may be involved in forming, executing, examining, or reassessing that decision.

Together, the artifacts provide three distinct perspectives:

IGA™ → WHERE the governance problem belongs.

GIARM™ → HOW governability persists through intelligent activity.

Governable Intelligent Activity Decision Tree™ → UNDER WHAT GOVERNANCE CONDITIONS activity may proceed.

28. What IGA™ Defines

IGA™ defines:

where governance belongs, the primary governance spaces of intelligent and autonomous systems, what belongs within each space, how governance spaces differ, how governance spaces may interact, where cross-space boundaries must remain visible, how primary governance ownership can be distinguished from cross-architecture relevance, and how specialized governed domains may connect to the universal governance topology.

IGA™ provides a common reference language for navigating governance across intelligent systems.

29. What IGA™ Does Not Define

IGA™ does not prescribe:

a specific AI architecture, a specific model, a vendor or platform, a neural-network design, a mandatory technical implementation, a mandatory lifecycle, a single regulatory regime, a universal risk methodology, or identical governance controls for every system.

IGA™ identifies where governance belongs.

The applicable architecture, methodology, organization, authority, or governance system determines how the identified space is governed in practice.

30. Application Scope

IGA™ is designed as a universal governance reference map applicable across:

AI systems, autonomous agents, multi-agent systems, robotics, Human-AI environments, cognitive automation, intelligent infrastructure, autonomous organizations, machine-to-machine environments, and future intelligent systems.

Its topology is intentionally technology-neutral so that new forms of intelligence can be incorporated without requiring the fundamental governance vocabulary to depend on a particular contemporary implementation.

31. Evolution of the Map

IGA™ is designed to evolve.

A new technology does not automatically justify a new governance space.

A new governance space should be recognized only when a genuinely distinct governed object, governance question, boundary, and governance purpose can be established.

This allows IGA™ to expand as intelligent systems evolve while preserving architectural discipline.

The map should therefore change when governance discovery requires it — not merely when technology terminology changes.

32. Master Reference Principle

IGA™ identifies where governance belongs before deeper governance begins.

It does not replace the governance architectures, reference models, standards, mechanisms, or domain-specific methodologies that govern particular spaces.

It provides the master diagnostic reference through which those spaces can be distinguished and navigated.

33. Canonical Closing Statement

IGA™ — Intelligence Governance Architecture provides the OOF® Master Governance Space Map for intelligent and autonomous systems. It separates governance into distinct primary spaces — Governance, Identity, Computation, Cognition, Memory, Agency, Execution, Operational Reality, Observation, Evidence, Validation, Attribution, Accountability, and Auditability — while distinguishing the cross-cutting governance dimensions and specialized governed domains that may operate across them. IGA™ does not treat intelligent systems as a single governance problem and does not impose a mandatory lifecycle. It provides a common governance topology for identifying where a problem belongs, how governance spaces relate, and where governance ownership must remain distinct as intelligent systems become increasingly autonomous, interconnected, and operationally consequential.

IGA™ identifies where governance belongs. OOF® Governance Architectures define how that space is governed.

Governed Links

[IGA™ Architecture Map](#)

[GIARM™ — Governed Intelligent Activity Reference Model](#)

[Layers of Governed Intelligence](#)

[Governable Intelligent Activity Decision Tree™](#)

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